

A Trace-Free Slice of the Minor-Invariant Projection Question

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Source Question

Guerra and Raita ask in Question 5.17 of arXiv:1909.03923 for conditions on a projection

$$T : \mathbb{R}^{m \times n} \rightarrow V \subseteq \mathbb{R}^{m \times n}$$

under which one can find a nonzero linear combination H_s of s -minors, $2 \leq s < \min\{m, n\}$, satisfying

$$H_s(X) = H_s(X + Y) \quad \text{for every } X \text{ and every } Y \text{ with } T(Y) = 0.$$

Equivalently, H_s is a linear combination of s -minors depending only on the coordinates of V . The source passage is shown in Figure 1.

This packet solves the natural trace-free square-matrix slice of the question.

Definitions

For subsets $I, J \subset \{1, \dots, n\}$ with $|I| = |J| = s$, write

$$M_{I,J}(X) = \det X_{I,J},$$

where $X_{I,J}$ is the submatrix with rows I and columns J , both listed in increasing order. Every linear combination of s -minors on $\mathbb{R}^{n \times n}$ has the form

$$H(X) = \sum_{|I|=|J|=s} c_{I,J} M_{I,J}(X).$$

Let

$$T(X) = X - \frac{\operatorname{tr} X}{n} I_n.$$

Then T is the orthogonal projection onto the trace-free matrices and $\ker T = \mathbb{R} I_n$. Thus the source condition becomes

$$H(X + tI_n) = H(X) \quad \text{for all } X \in \mathbb{R}^{n \times n}, t \in \mathbb{R}.$$

Proof Intuition

If $2s \leq n$, there are disjoint row and column sets I, J of size s ; the minor $M_{I,J}$ does not involve any diagonal entry, so adding tI_n leaves it unchanged.

The converse is the useful part. If $2s > n$, every two s -subsets I, J meet. Differentiating the identity $H(X + tI_n) = H(X)$ in t gives a linear relation among $(s-1)$ -minors. The coefficient of a fixed $(s-1)$ -minor is a signed sum of the coefficients of the s -minors obtained by adding the same index to the row and column sets. A descending induction on $|I \cup J|$ then isolates and kills all coefficients.

Main Partial Theorem

Theorem 1 (Trace-free scalar-shift threshold). *Let $n \geq 3$ and $2 \leq s < n$. Let $T : \mathbb{R}^{n \times n} \rightarrow \{X : \text{tr } X = 0\}$ be*

$$T(X) = X - \frac{\text{tr } X}{n} I_n.$$

There is a nonzero linear combination H of s -minors such that

$$H(X) = H(X + Y) \quad \text{whenever } T(Y) = 0$$

if and only if

$$2s \leq n.$$

Proof. Since $\ker T = \mathbb{R}I_n$, the invariance condition is exactly

$$H(X + tI_n) = H(X) \quad \text{for all } X, t.$$

If $2s \leq n$, choose disjoint subsets $I, J \subset \{1, \dots, n\}$ with $|I| = |J| = s$. For $i \in I$ and $j \in J$ we have $i \neq j$, so the entries of the submatrix $X_{I,J}$ are unchanged by adding tI_n . Hence $H = M_{I,J}$ is a nonzero invariant linear combination of s -minors.

Conversely, assume $2s > n$ and let

$$H(X) = \sum_{|I|=|J|=s} c_{I,J} M_{I,J}(X)$$

be invariant. Differentiating at $t = 0$ gives

$$0 = \left. \frac{d}{dt} \right|_{t=0} H(X + tI_n)$$

as a polynomial identity in X . For a single minor,

$$\left. \frac{d}{dt} \right|_{t=0} M_{I,J}(X + tI_n) = \sum_{a \in I \cap J} \varepsilon(a; I, J) M_{I \setminus \{a\}, J \setminus \{a\}}(X),$$

where each $\varepsilon(a; I, J)$ is 1 or -1 . The precise sign will not matter.

The minors of a fixed order are linearly independent as polynomials, since a monomial occurring in $M_{A,B}$ uses exactly the row set A and the column set B . Therefore, for every pair A, B of $(s-1)$ -subsets, the coefficient of $M_{A,B}$ in the differentiated identity is zero:

$$\sum_{a \notin A \cup B} \varepsilon(a; A \cup \{a\}, B \cup \{a\}) c_{A \cup \{a\}, B \cup \{a\}} = 0. \quad (1)$$

We prove that all coefficients $c_{I,J}$ vanish by descending induction on $N = |I \cup J|$. Since $2s > n$, every pair of s -sets I, J has $I \cap J \neq \emptyset$.

Start with $N = n$. Pick $a \in I \cap J$ and set $A = I \setminus \{a\}$, $B = J \setminus \{a\}$. Then $A \cup B = (I \cup J) \setminus \{a\}$ has complement exactly $\{a\}$. Thus (1) contains only the coefficient $c_{I,J}$, up to a nonzero sign, and so $c_{I,J} = 0$.

Now suppose all coefficients with union size larger than N vanish, and take I, J with $|I \cup J| = N$. Choose $a \in I \cap J$ and set $A = I \setminus \{a\}$, $B = J \setminus \{a\}$. In (1), the term with the added index a is precisely $c_{I,J}$, up to sign. Any other possible added index $b \notin A \cup B$ lies outside $I \cup J$, and the corresponding coefficient $c_{A \cup \{b\}, B \cup \{b\}}$ has union size $N + 1$; it is zero by the induction hypothesis. Hence $c_{I,J} = 0$.

The descending induction reaches every pair I, J , so all coefficients vanish. Thus no nonzero invariant combination exists when $2s > n$. $\square$

Scope and Novelty Check

The theorem gives an exact existence threshold for the projection onto trace-free square matrices. It is a partial answer to Question 5.17, not a classification for arbitrary projections T . In the positive range $2s \leq n$, the theorem gives explicit invariants but does not describe the entire invariant subspace.

Cheap run indexes were searched for arXiv:1909.03923, trace-free, traceless, scalar-shift, identity direction, minors, and the source question keywords. No duplicate packet was found. A bounded web search on July 19, 2026 found the source paper and preprint records but no exact answer to this trace-free slice.

Computational Sanity Check

The script `code/check_tracefree_threshold.py` builds the exact rational matrix of the identity-direction derivative on s -minors and checks the kernel dimensions for $3 \leq n \leq 7$. It confirms the threshold above for the tested cases. This computation is only a verification aid; the proof above is independent.

References

1. A. Guerra and B. Raita, *Quasiconvexity, null Lagrangians, and Hardy space integrability under constant rank constraints*, arXiv:1909.03923; Archive for Rational Mechanics and Analysis 245 (2022), 279–320.

where $T \equiv P_{\text{asym}}$ denotes the orthogonal projection onto anti-symmetric matrices. We will test the relation $H_2(X) = H_2(T(X))$ with different matrices $X \in \mathbb{R}^{3 \times 3}$ to show that $H_2 = 0$, since this is enough to show that there are no non-affine null Lagrangians (see also Example 5.14). First, note that by taking $X = e_i \otimes e_i + e_j \otimes e_j$, $i \neq j$, the coefficients of the principal minors in H_2 must be zero. The other 2×2 minors touch the main diagonal on exactly one entry, say (i, i) . By taking $X = ae_i \otimes e_i + e_j \otimes e_k$, $j \neq i \neq k$, for $a \in \mathbb{R}$, we see that indeed $H_2 = 0$.

For general dimension $n \geq 3$, it is not too difficult to see that there are no non-affine div-null Lagrangians.

It would be interesting to give a theoretical characterization of the solutions of the computational problem. This is also a relevant question since the linear system described above grows factorially in $\dim \mathbb{V}$, although in applications to continuum mechanics this number is usually relatively small. Unfortunately, even in the special case when $\mathcal{B}$ has order one such a characterization seems difficult. The authors were unable to give a definitive answer even to the following simple-looking question.

Assume we are given a projection $T: \mathbb{R}^{m \times n} \rightarrow \mathbb{V}$, which can be chosen to be orthogonal, onto some subspace $\mathbb{V} \subseteq \mathbb{R}^{m \times n}$. Consider a function $H_s: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ as above, i.e.

$$H_s(X) = \sum_{\deg M=s} c_M M(X), \quad H_s = H_s \circ T$$

where the sum runs over the set of all minors (not necessarily principal) of order $2 \leq s < \min\{m, n\}$. The second condition can be equivalently rewritten as

$$(5.16) \quad H_s(X) = H_s(X + Y) \text{ for all } X, \text{ all } Y \text{ such that } T(Y) = 0.$$

We think of this as saying that the linear combination of minors H_s only depends on the coordinates of $\mathbb{V}$.

5.17 Question. Under which conditions on T can we find non-zero H_s satisfying (5.16)? Can we characterize such H_s in terms of $\mathbb{V}$?

6 Compensated compactness in Hardy spaces

We begin by stating the main theorem of this section; as usual, we assume (1.5) holds throughout. Recall that A -quasiaffine maps are polynomials (c.f. Proposition 5.3) and see Definition 3.9 for the definition of $\mathbb{I}_A$.

6.1 Theorem. *Let $F: \mathbb{V} \rightarrow \mathbb{R}$ be locally bounded and Borel. If the implication*

$$(6.2) \quad v \in C_{c,A}^\infty(\mathbb{R}^n) \implies F(v) \in \mathcal{H}^1(\mathbb{R}^n)$$

holds, F is a sum of homogeneous A -quasiaffine functions of degree at most $\min\{n, \dim \mathbb{V}\}$.

Conversely, assume that F is an s -homogeneous A -quasiaffine function. If $s \geq 2$ then (6.2) holds and moreover

$$\|F(v)\|_{\mathcal{H}^1} \leq C \|v\|_{L^p}^s \quad \text{for } v \in C_{c,A}^\infty(\mathbb{R}^n).$$

If $s = 1$, we have that $F(v) = v_0 \cdot v$ for some $v_0 \in \mathbb{V}$ and (6.2) holds if and only if $v_0 \perp \mathbb{I}_A$, although nonetheless the above estimate fails.

Figure 1: Question 5.17 in the source paper, printed page 24.